

Kfre: An R Package for Kidney Failure Risk Estimation Using the Kidney Failure Risk Equation

by Leonid Shpaner

Abstract Chronic kidney disease (CKD) affects hundreds of millions of people worldwide, and identifying patients at highest risk of progression to kidney failure is critical for timely clinical intervention. The Kidney Failure Risk Equation (KFRE), developed by Tangri and colleagues and validated across more than 700,000 individuals in 30 countries, provides 2- and 5-year probability estimates of kidney failure using 4, 6, or 8 clinical variables. Despite its widespread clinical adoption and endorsement by KDIGO guidelines, no dedicated, feature-complete R implementation existed prior to this work. The `kfre` package fills this gap by providing vectorized KFRE predictions across all three model variants and both time horizons, a convenience wrapper for appending risk columns to data frames, CKD stage and ESRD outcome classification utilities, urine protein-to-creatinine ratio conversion, and publication-ready ROC and precision-recall curve plotting. The package is designed with three complementary interfaces: a functional API, a data-frame wrapper, and an R6 class, to serve clinicians, analysts, and pipeline developers alike. This paper describes the package design, implementation, and a fully reproducible demonstration using synthetic patient data. This package is available from the Comprehensive R Archive Network at <https://CRAN.R-project.org/package=kfre>

1 Introduction

Chronic kidney disease (CKD) is a major global health burden, affecting an estimated 10-15% of adults worldwide (Bikbov et al., 2020). For the subset of patients whose disease progresses to kidney failure, requiring dialysis or transplantation, the clinical and economic consequences are severe. Identifying these high-risk individuals early enough to intervene meaningfully is one of the central challenges of nephrology practice.

The Kidney Failure Risk Equation (KFRE) was developed by Tangri et al. (2011) using data from two Canadian cohorts of patients with CKD stages G3-G5, and subsequently validated by Tangri et al. (2016) in over 700,000 individuals spanning more than 30 countries. The KFRE provides individualized 2- and 5-year probability estimates of progression to kidney failure using either four variables (age, sex, estimated glomerular filtration rate [eGFR], and urine albumin-to-creatinine ratio [uACR]), six variables (adding diabetes and hypertension status), or eight variables (additionally incorporating serum albumin, phosphorus, bicarbonate, and calcium). It has since been incorporated into electronic health record systems, reimbursement criteria, and international clinical practice guidelines. Specifically, the Kidney Disease: Improving Global Outcomes (KDIGO) 2024 guidelines recommend KFRE as the standard tool for estimating absolute kidney failure risk in patients with CKD stages G3-G5 (Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group, 2024).

Despite this broad clinical adoption, researchers and data analysts working in R lacked a dedicated, full-featured package for computing KFRE estimates, classifying CKD staging, converting protein measurement units, and evaluating model performance, all within a single, coherent interface. Prior to `kfre`, analysts typically implemented the equations manually or adapted code from other languages, introducing opportunities for transcription errors and inconsistencies across projects. The `kfre` package (Shpaner, 2025) addresses this gap.

A companion Python implementation exists (available via PyPI), and `kfre` is designed to maintain parity with it in naming conventions, output shapes, and expected column structures, facilitating cross-language reproducibility in research pipelines that use both R and Python.

The remainder of this paper is organized as follows. We describe the KFRE methodology in Section 2, the package design and architecture in Section 3, each major component of the API in Section 4, a reproducible worked example in Section 5, and concluding discussion in Section 6.

Table 1: Published KFRE log-hazard coefficients from Tangri et al. (2016) for North American populations.

Variable	4-variable	6-variable	8-variable
Age (per year)	-0.2201	-0.2201	-0.1992
Sex (male = 1)	-0.2467	-0.2467	-0.3580
eGFR (per mL/min/1.73 m ²)	-0.5567	-0.5567	-0.4510
ln(uACR) (per unit)	0.4510	0.4510	0.3770
Diabetes mellitus	—	0.2097	0.1870
Hypertension	—	0.1031	0.1430
Albumin (g/dL)	—	—	-0.3010
Phosphorus (mg/dL)	—	—	0.2030
Bicarbonate (mEq/L)	—	—	-0.1360
Calcium (mg/dL)	—	—	-0.1830

2 Background

The KFRE risk model

The KFRE models the probability of kidney failure within a given time horizon using a proportional hazards framework. For a patient with covariate vector $\mathbf{x}$, the predicted probability of kidney failure within t years is:

$$P(\text{kidney failure within } t \text{ years} \mid \mathbf{x}) = 1 - S_0(t)^{\exp(\mathbf{x}^T \boldsymbol{\beta} - c)}$$

where $S_0(t)$ is the baseline survival probability at horizon t , $\boldsymbol{\beta}$ is the vector of published log-hazard coefficients, and c is a centering constant equal to the mean linear predictor in the derivation cohort. The term $\exp(\mathbf{x}^T \boldsymbol{\beta} - c)$ is the relative hazard for the individual patient relative to the average patient in the derivation sample.

The linear predictor for the 4-variable model takes the form:

$$\eta_4 = \beta_{\text{age}} \cdot \text{age} + \beta_{\text{sex}} \cdot \mathbf{1}[\text{male}] + \beta_{\text{eGFR}} \cdot \text{eGFR} + \beta_{\text{uACR}} \cdot \ln(\text{uACR})$$

The 6-variable model extends this with:

$$\eta_6 = \eta_4 + \beta_{\text{dm}} \cdot \mathbf{1}[\text{diabetes}] + \beta_{\text{htm}} \cdot \mathbf{1}[\text{hypertension}]$$

And the 8-variable model further adds:

$$\eta_8 = \eta_6 + \beta_{\text{alb}} \cdot \text{albumin} + \beta_{\text{phos}} \cdot \text{phosphorus} + \beta_{\text{bicarb}} \cdot \text{bicarbonate} + \beta_{\text{ca}} \cdot \text{calcium}$$

Note that uACR enters as its natural logarithm, reflecting the log-linear relationship between albuminuria and kidney failure risk observed across the derivation and validation cohorts.

Published coefficients

Table 1 presents the published coefficients from Tangri et al. (2016) for all three model variants. The coefficients are identical across the 2-year and 5-year horizons within each population stratum; only the baseline survival probability $S_0(t)$ differs by horizon.

Baseline survival probabilities

The baseline survival probabilities $S_0(t)$ differ by time horizon and by whether the North American or non-North American calibration is applied. Table 2 presents these values as reported by Tangri et al. (2016). The non-North American values are systematically lower, reflecting higher observed event rates in international cohorts relative to the Canadian derivation sample.

The non-North American calibration applies a multiplicative shift to the baseline survival. If $S_0^{\text{NA}}(t)$ denotes the North American baseline survival and $S_0^{\text{non-NA}}(t)$ the international value, the

Table 2: Baseline survival probabilities $S_0(t)$ by model, time horizon, and population.

Model	Horizon	North American	Non-North American
4-variable	2-year	0.9832	0.9766
4-variable	5-year	0.9365	0.9240
6-variable	2-year	0.9832	0.9766
6-variable	5-year	0.9365	0.9240
8-variable	2-year	0.9769	0.9695
8-variable	5-year	0.9240	0.9113

recalibrated probability for a patient with linear predictor η is simply:

$$p^{\text{non-NA}}(t | \mathbf{x}) = 1 - \left[S_0^{\text{non-NA}}(t) \right]^{\exp(\eta - c)}$$

This recalibration adjusts the overall event rate while preserving the relative ranking of patients by risk, since the coefficient vector β is held constant across populations.

uPCR to uACR conversion

The KFRE requires uACR as input, but many clinical datasets record only urine protein-to-creatinine ratio (uPCR). **kfre** implements the sex-, diabetes-, and hypertension-stratified log-linear regression equations from [Sumida et al. \(2020\)](#). The general form is:

$$\ln(\hat{\text{uACR}}) = \alpha_k + \beta_k \cdot \ln(\text{uPCR})$$

where k indexes the subgroup defined by sex, diabetes status, and hypertension status, and α_k, β_k are the subgroup-specific intercept and slope estimated by [Sumida et al. \(2020\)](#) from a large multi-ethnic CKD cohort. Exponentiating gives the predicted uACR on the original scale:

$$\hat{\text{uACR}} = \exp(\alpha_k + \beta_k \cdot \ln(\text{uPCR}))$$

This approach produces better-calibrated uACR estimates than a simple fixed ratio because the uPCR-to-uACR relationship varies systematically with patient characteristics: diabetic patients tend to excrete proportionally more albumin relative to total protein, and females tend to have lower creatinine excretion, both of which affect the ratio between the two measurements.

3 Package design

Design philosophy

kfre is designed around three complementary usage patterns that reflect distinct user types and workflows in clinical research:

Single-person queries are served by the `kfre_person()` function (also available as a method on the `RiskPredictor R6` object). This interface accepts individual scalar values and is well-suited for interactive exploration, clinical decision support prototyping, or embedding in Shiny applications.

Cohort-level analysis is served by `add_kfre_risk_col()`, a convenience wrapper that accepts a data frame and column name mappings and returns the same data frame augmented with one new column per requested model-and-horizon combination. This follows the common R idiom of returning an enriched data frame and fits naturally into **dplyr**-style analysis pipelines.

Programmatic and pipeline use is served by the `RiskPredictor R6` class, which encapsulates the data frame and column mappings at construction time and exposes a `predict_kfre()` method for vectorized prediction. This design is particularly useful for developers building larger clinical data pipelines where the predictor object is passed between functions or modules, and for users coming from Python who are familiar with scikit-learn-style `fit/predict` patterns.

This multi-interface approach means that a nephrologist testing a single patient, a biostatistician processing a registry cohort, and a software engineer building an EHR plugin can all use **kfre** via the interface most natural to their context.

Dependencies

kfre imports **R6** (Chang et al., 2021) for the object-oriented interface, **ggplot2** (Wickham, 2016) for plotting, **pROC** (Robin et al., 2011) for ROC curve computation, and **precrec** (Saito and Rehmsmeier, 2015) for precision-recall curve computation. The choice of **pROC** and **precrec** in preference to computing metrics manually was deliberate: these packages implement partial AUC, confidence intervals, and bootstrapping that would otherwise require substantial additional code, and they maintain parity with the metric implementations in the companion Python package.

Object model

The `RiskPredictor` class uses the R6 object model. The R6 approach was chosen over S3 or S4 because the class encapsulates both data state (the input data frame and column mapping) and behavior (the prediction methods) in a way that is more natural for users constructing a predictor once and calling it multiple times. This also simplifies the public API by reducing the number of arguments that must be repeated across multiple calls.

The functional wrappers `predict_kfre()` and `add_kfre_risk_col()` are thin convenience layers over the same underlying core function `risk_pred_core()`, ensuring that all interfaces share a single implementation and that any bug fix propagates to all entry points.

4 Package components

Core prediction: `risk_pred_core()`

`risk_pred_core()` is the internal computation engine. It accepts scalar or vector inputs (age, sex indicator, eGFR, uACR, and optionally the extended variables), the desired number of model variables (4, 6, or 8), the prediction horizon (2 or 5 years), and a logical flag for North American vs. non-North American coefficients. It returns a numeric vector of predicted probabilities. Direct use of `risk_pred_core()` is not required by most users; it is called internally by the higher-level interfaces.

Single-person prediction: `kfre_person()`

`kfre_person()` provides a named-argument interface for computing the KFRE for a single patient. This function is intended to be readable at a glance:

```
# 4-variable KFRE, 2-year horizon, male patient
rp <- RiskPredictor$new(df = toy, columns = cols)

rp$kfre_person(
  age = 55, is_male = TRUE,
  eGFR = 45, uACR = 120,
  is_north_american = TRUE, years = 2
)
```

The `is_male` argument (rather than a string) was chosen to make the binary nature of the sex input explicit and to avoid ambiguity over string matching (e.g., "M" vs. "male" vs. "Male"). The `add_kfre_risk_col()` wrapper accepts a column of character sex labels and handles normalization internally via `perform_conversions()`.

Data-frame wrapper: `add_kfre_risk_col()`

For cohort analysis, `add_kfre_risk_col()` accepts a data frame along with named arguments mapping KFRE variable names to column names in the data frame. It returns the data frame with additional columns appended, named using the convention `kfre_{n}var_{t}year` (e.g., `kfre_4var_2year`). The `num_vars` and `years` arguments each accept vectors, so all desired combinations can be computed in a single call:

```
toy <- data.frame(
  age      = c(55, 72),
  sex_txt  = c("male", "female"),
```

Table 3: KFRE risk estimates: 4- and 6-variable models.

Age	Sex	eGFR	uACR	4v-2yr	4v-5yr	6v-2yr	6v-5yr
55	male	45	120	0.0125	0.0384	0.0120	0.0369
72	female	28	800	0.1000	0.2803	0.1094	0.3036

Table 4: KFRE risk estimates: 8-variable model.

Age	Sex	8v-2yr	8v-5yr
55	male	0.0113	0.0362
72	female	0.1193	0.3389

```

eGFR      = c(45, 28),
uACR      = c(120, 800),
dm        = c(1, 0),
htn       = c(1, 1),
albumin   = c(4.2, 3.4),
phosphorous = c(3.3, 4.6),
bicarbonate = c(24, 22),
calcium    = c(9.1, 9.8),
stringsAsFactors = FALSE
)

toy_kfre <- add_kfre_risk_col(
  df          = toy,
  age_col     = "age",
  sex_col     = "sex_txt",
  eGFR_col    = "eGFR",
  uACR_col    = "uACR",
  dm_col      = "dm",
  htn_col     = "htn",
  albumin_col = "albumin",
  phosphorous_col = "phosphorous",
  bicarbonate_col = "bicarbonate",
  calcium_col = "calcium",
  num_vars    = c(4, 6, 8),
  years       = c(2, 5),
  is_north_american = TRUE,
  copy        = TRUE
)

```

Tables 3 and 4 show predicted risk estimates for both patients across the 4- and 6-variable models and the 8-variable model respectively.

The `copy = TRUE` argument returns a new data frame rather than modifying the input in place, consistent with R's functional programming conventions and predictable behavior in pipeline contexts.

R6 interface: RiskPredictor

The `RiskPredictor` R6 class encapsulates a data frame and its column mappings at construction time, exposing `predict_kfre()` for vectorized batch prediction and `kfre_person()` for single-patient queries. This interface is particularly useful when predictions across multiple model configurations are needed repeatedly on the same dataset, as the data frame and column map need only be specified once:

```

cols <- list(
  age = "age", sex = "sex_txt", eGFR = "eGFR", uACR = "uACR",
  dm = "dm", htn = "htn", albumin = "albumin",
  phosphorous = "phosphorous", bicarbonate = "bicarbonate",
  calcium = "calcium"
)

```

Table 5: Predictions via RiskPredictor R6 interface.

Patient	4v-2yr	6v-5yr	8v-2yr
Patient 1 (M, 55)	0.0125	0.0369	0.0113
Patient 2 (F, 72)	0.1000	0.3036	0.1193

```
rp <- RiskPredictor$new(df = toy, columns = cols)
p4_2y <- rp$predict_kfre(years = 2, is_north_american = TRUE,
  use_extra_vars = FALSE, num_vars = 4)
p6_5y <- rp$predict_kfre(years = 5, is_north_american = TRUE,
  use_extra_vars = TRUE, num_vars = 6)
p8_2y <- rp$predict_kfre(years = 2, is_north_american = TRUE,
  use_extra_vars = TRUE, num_vars = 8)
```

CKD staging and ESRD outcome classification

`class_ckd_stages()` assigns standard CKD stage labels (G1-G5, with G3 split into G3a and G3b) based on eGFR, following KDIGO classification thresholds. An optional `combined_stage_col` merges adjacent stages for analyses that use broader groupings (e.g., G3a/G3b combined as G3).

`class_esrd_outcome()` constructs binary outcome columns from a continuous follow-up duration and an ESRD event flag, for use as ground-truth labels in model evaluation. It accepts follow-up time in days (converting internally to years) and a threshold horizon, returning a binary column indicating whether the ESRD event occurred within the specified window:

```
out <- data.frame(
  eGFR = c(65, 48, 25, 12),
  ESRD_flag = c(0, 1, 1, 1),
  followup_days = c(400, 600, 500, 800)
)

out <- class_ckd_stages(
  df = out, egfr_col = "eGFR",
  stage_col = "stage", combined_stage_col = "stage_combined"
)

out <- class_esrd_outcome(
  df = out, col = "ESRD_flag", years = 2,
  duration_col = "followup_days", prefix = "esrd",
  create_years_col = TRUE
)
```

Applied to four synthetic patients with eGFR values of 65, 48, 25, and 12 mL/min/1.73 m², `class_ckd_stages()` assigns stages G2, G3a, G4, and G5 respectively. `class_esrd_outcome()` then labels the 2-year binary outcome: patients with eGFR 48 and 25 (follow-up of 600 and 500 days, both under 2 years) are flagged as events, while the patient with eGFR 65 (follow-up 400 days, no ESRD flag) and the patient with eGFR 12 (follow-up 800 days, exceeding the 2-year window) are labeled 0.

uPCR to uACR conversion: `upcr_uacr()`

The KFRE requires uACR, but many datasets record urine protein-to-creatinine ratio (uPCR). `kfre` implements the sex-, diabetes-, and hypertension-stratified conversion equations from [Sumida et al. \(2020\)](#), which provide better calibration for clinical populations than a simple fixed ratio:

```
df_upcr <- data.frame(
  sex = c("female", "male", "female"),
  dm = c(1, 0, 1),
  htn = c(1, 1, 0),
  upcr = c(150, 600, 50)
```

```
)

uacr <- upcr_uacr(
  df_upcr,
  sex_col      = "sex",
  diabetes_col  = "dm",
  hypertension_col = "htn",
  upcr_col     = "upcr",
  female_str    = "female"
)
```

The three input uPCR values (150, 600, and 50 mg/g) convert to estimated uACR values of 33.5, 269.06, and 5.29 mg/g respectively, reflecting the sex-, diabetes-, and hypertension-stratified adjustments applied by the [Sumida et al. \(2020\)](#) regression equations.

Model evaluation: `eval_kfre_metrics()`

`eval_kfre_metrics()` computes a suite of performance metrics for each requested model variant and time horizon. Three metrics are implemented.

The **Brier score** measures mean squared error between predicted probabilities $\hat{p}_i$ and binary outcomes $y_i \in \{0, 1\}$ over n patients:

$$BS = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2$$

A Brier score of 0 indicates perfect calibration; a score of 0.25 corresponds to the performance of an uninformative model that predicts $\hat{p}_i = 0.5$ for all patients.

The **area under the ROC curve** (AUC-ROC) measures the probability that a randomly chosen event patient receives a higher predicted risk than a randomly chosen non-event patient:

$$AUC = P(\hat{p}_i > \hat{p}_j \mid y_i = 1, y_j = 0)$$

An AUC of 0.5 corresponds to random discrimination; an AUC of 1.0 to perfect separation of events from non-events.

The **average precision** (AP) summarizes the precision-recall curve as the weighted mean of precisions at each threshold, with the increase in recall from the previous threshold used as the weight:

$$AP = \sum_k (R_k - R_{k-1}) \cdot P_k$$

where P_k and R_k are the precision and recall at the k -th threshold. Average precision is particularly informative for imbalanced datasets where kidney failure events are rare relative to the total cohort, as it penalizes models that achieve high recall only at the cost of many false positives.

`eval_kfre_metrics()` expects outcome columns following the naming convention of `class_esrd_outcome()` (e.g., `esrd_2_year_outcome`) and prediction columns following that of `add_kfre_risk_col()` (e.g., `kfre_4var_2year`). This convention is enforced consistently so that evaluation requires no additional column renaming.

Visualization: `plot_kfre_metrics()`

`plot_kfre_metrics()` produces publication-quality ROC and precision-recall curves using **ggplot2**. It supports overlaying multiple model variants on the same axes (useful for comparing the 4-, 6-, and 8-variable models), and can optionally save plots to disk in PNG or SVG format. The `plot_type` argument accepts `"auc_roc"`, `"pr_curve"`, or `"all_plots"`, and `mode` controls whether to compute metrics, plot from pre-computed results, or do both.

5 Worked example

To demonstrate the full package workflow end to end, we construct a synthetic dataset of 200 patients with CKD, compute KFRE predictions, assign CKD stages and ESRD outcome labels, and visualize the distribution of predicted risk across CKD stages.

Synthetic data

```
set.seed(42)
n <- 200

synth <- data.frame(
  age      = round(runif(n, 40, 85)),
  sex_txt  = sample(c("male", "female"), n, replace = TRUE),
  eGFR     = round(runif(n, 10, 59)),
  uACR     = round(runif(n, 30, 3000)),
  dm       = sample(0:1, n, replace = TRUE, prob = c(0.4, 0.6)),
  htn      = sample(0:1, n, replace = TRUE, prob = c(0.25, 0.75)),
  albumin  = round(runif(n, 2.5, 5.0), 1),
  phosphorous = round(runif(n, 2.5, 6.5), 1),
  bicarbonate = round(runif(n, 18, 28)),
  calcium  = round(runif(n, 8.0, 10.5), 1),
  ESRD_flag = sample(0:1, n, replace = TRUE, prob = c(0.6, 0.4)),
  followup_days = round(runif(n, 180, 1825)),
  stringsAsFactors = FALSE
)
```

Computing KFRE and staging

```
synth_kfre <- add_kfre_risk_col(
  df = synth, age_col = "age", sex_col = "sex_txt",
  eGFR_col = "eGFR", uACR_col = "uACR",
  dm_col = "dm", htn_col = "htn",
  albumin_col = "albumin", phosphorous_col = "phosphorous",
  bicarbonate_col = "bicarbonate", calcium_col = "calcium",
  num_vars = c(4, 6, 8), years = c(2, 5),
  is_north_american = TRUE, copy = TRUE
)

synth_kfre <- class_ckd_stages(
  df = synth_kfre, egfr_col = "eGFR",
  stage_col = "stage", combined_stage_col = "stage_combined"
)

synth_kfre <- class_esrd_outcome(
  df = synth_kfre, col = "ESRD_flag", years = 2,
  duration_col = "followup_days", prefix = "esrd",
  create_years_col = TRUE
)

synth_kfre <- class_esrd_outcome(
  df = synth_kfre, col = "ESRD_flag", years = 5,
  duration_col = "followup_days", prefix = "esrd",
  create_years_col = FALSE
)
```

Risk distribution by CKD stage

The distribution of 4-variable, 2-year KFRE estimates by CKD stage confirms the expected relationship: patients in more advanced stages carry substantially higher predicted risk (Figure 1).

Comparing model variants

The 4- and 8-variable models produce closely correlated estimates (Figure 2), with systematic divergence at higher risk levels reflecting the additional information contributed by the extended laboratory variables.

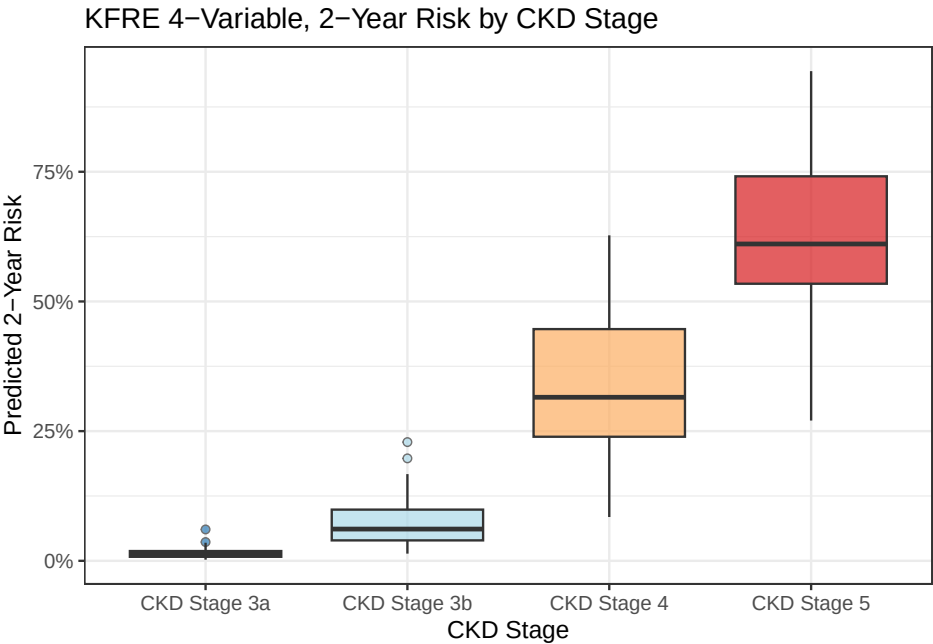


Figure 1: Distribution of 4-variable, 2-year KFRE predicted risk by CKD stage in synthetic cohort.

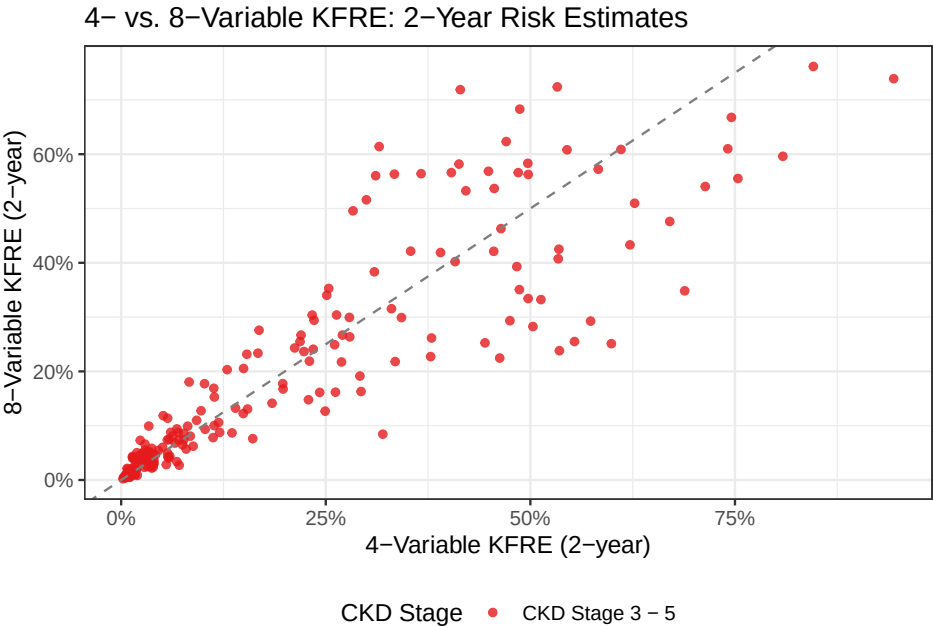


Figure 2: Predicted 2-year risk: 4-variable vs. 8-variable KFRE in synthetic cohort.

6 Discussion

kfre provides the R community with a comprehensive, tested, and reproducible implementation of the Kidney Failure Risk Equation. The package addresses a practical gap: the KFRE is now a guideline-recommended clinical tool used globally, yet analysts working in R were previously required to implement the equations manually or adapt code from other sources. **kfre** consolidates the full analytical workflow from raw data processing through risk prediction, staging, evaluation, and visualization into a single, cohesive package.

Several design decisions merit explicit discussion. First, the three-interface architecture (functional, data-frame wrapper, R6) is unusual but deliberate. A single-function approach would have forced trade-offs between ease of use for interactive work and ergonomics for pipeline development; the three interfaces address these use cases without duplicating logic. Second, the naming convention for output columns (`kfre_{n}var_{t}year`) is enforced consistently across `add_kfre_risk_col()`, `eval_kfre_metrics()`, and `plot_kfre_metrics()`, so that the output of one function can be passed directly to the next without renaming.

Limitations. The KFRE coefficients implemented in **kfre** are those published by Tangri et al. (2011) and Tangri et al. (2016). Recent work, including the 59-cohort CKD Prognosis Consortium study using the 2021 CKD-EPI creatinine equation (Grams et al., 2023), has evaluated the KFRE with updated eGFR-estimating equations and found generally good performance for $\text{eGFR} < 45 \text{ mL/min/1.73 m}^2$. Users applying **kfre** in populations with eGFR between 45 and 60, or in settings where competing risks from all-cause mortality are substantial, should consult this literature before interpreting predictions. Second, the non-North American calibration factor shifts the baseline survival probability for non-North American cohorts, but users should verify calibration in their own population before clinical deployment. Third, the uPCR-to-uACR conversion from Sumida et al. (2020) introduces its own measurement uncertainty; direct uACR measurement remains preferable when available.

Future directions. Planned extensions include support for updated coefficient sets as they are published, confidence interval estimation via bootstrapping, and a Shiny-based interface for single-patient point-of-care use. Contributions and bug reports are welcomed via the GitHub issue tracker at https://github.com/lshpaner/kfre_r/issues.

kfre is available on CRAN and is distributed under the MIT License.

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